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People's Democratic Republic of Algeria

Ministry of Higher Education and Scientific Research

University of Batna 2 Mostapha Benboulaid





Faculty of Mathematics and Computer Science

Department of Computer Science

Bachelor's Thesis

Specialization: Computer Systems

Topic

DESIGN AND DEVELOPMENT OF A WEB-BASED DOCTOR MANAGEMENT APPLICATION

Presented by:

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Class of: 2018/2019

Dedicatio ns

To:

My dear parents. My family.

My friends.

Everyone who loves me and whom I love.

Bilal

Dedications

To:

My dear parents. My

family.

My friends.

Everyone who loves me and whom I love.

Mohammed El Amine

Acknowledgments

We would like to express our gratitude to:

Our advisor, Mr. Sedrati, and Mr. Betta
We would also like to express our gratitude to him for his patience and support, which were invaluable in helping us successfully complete our work.

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General Introduction

In computer science, a web application is an application that can be used directly online via a web browser and therefore does not require installation on client machines, unlike mobile or desktop applications. Similar to websites, a web application is typically hosted on a server and is operated by interacting with widgets using a web browser over a computer network.

Today, clinics in Algeria face challenges in managing their operations and ensuring they run smoothly.

Since these clinics do not have an application that allows them to manage appointments or patient records, doctors and their assistants find themselves overwhelmed with tasks that could easily be simplified.

In this case, you must create a web application to take advantage of the many services provided by the web.

This thesis is organized into three chapters:

The first chapter defines the requirements for our application and outlines the requirements specification using a use-case diagram.

The second chapter presents the application design using UML. It includes the modeling diagrams.

The third chapter presents the various tools used for implementation as well as all the interfaces.

Chapter I : Definitions and Requirements Analysis

I.1 Introduction

The purpose of this chapter is to define and specify the requirements for a web application designed to manage patient appointments with a doctor or clinic. After defining the requirements for our application, we will use the use-case diagram, which is one of the UML diagrams.

I.2 Definition of Requirements

People generally prefer to find what they're looking for easily, especially when it comes to the medical field; that's why we've decided to offer a web application that allows users to find and contact a doctor and schedule an appointment.

Doctor Management:

This must include:

- Adding, deleting, modifying, and viewing a doctor's schedule.

Patient Management:

It must provide:

- Add, delete, edit, and view patients.

Appointment management:

It must provide the following:

- Adding, deleting, editing, and viewing appointments.

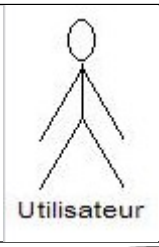
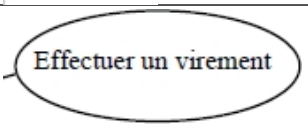
The app allows patients to find a doctor to schedule an appointment and view useful information. Additionally, our app serves as an advertising platform for the doctor's services. Meanwhile, the doctor or clinic manages patients using the app.

I.3 Use Case Diagrams

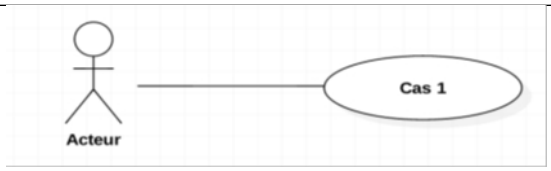
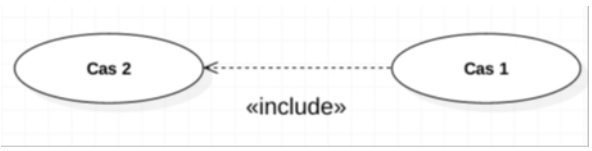
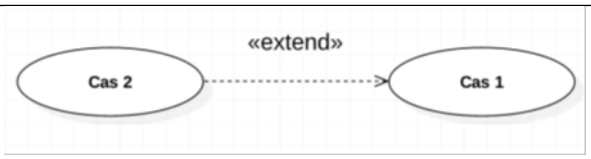
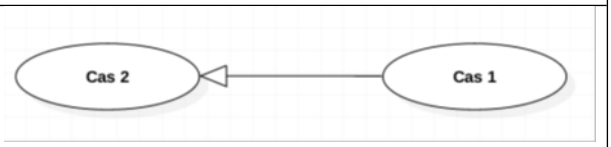
A use case diagram is an actor diagram that includes a set of use cases within the system boundaries, communication (or association) relationships between actors and use cases, and generalizations of these use cases. [1]

I.3.1 Notations

I.3.1.1 Elements of use case diagrams:

Element	Description	Graphical representation
Actor	An actor represents a role played by an external entity (human user, hardware device, or another system) that interacts directly with the system under study.	
Use Case	A use case represents a set of sequences of actions performed by the system that produce an observable result of interest to a particular actor.	

I.3.1.2 Relationships in Use Case Diagrams

Type	Description	Graphical Representation
The association	Is a communication link between an actor and a use case.	
Inclusion	Specifies that a use case is necessarily part of another use case.	
Extension	Specifies that a use case may be a part of another use case.	
Generalization	Is the application of the concept of inheritance in the object-oriented paradigm to use cases.	

I.3.2 Use Case Diagram

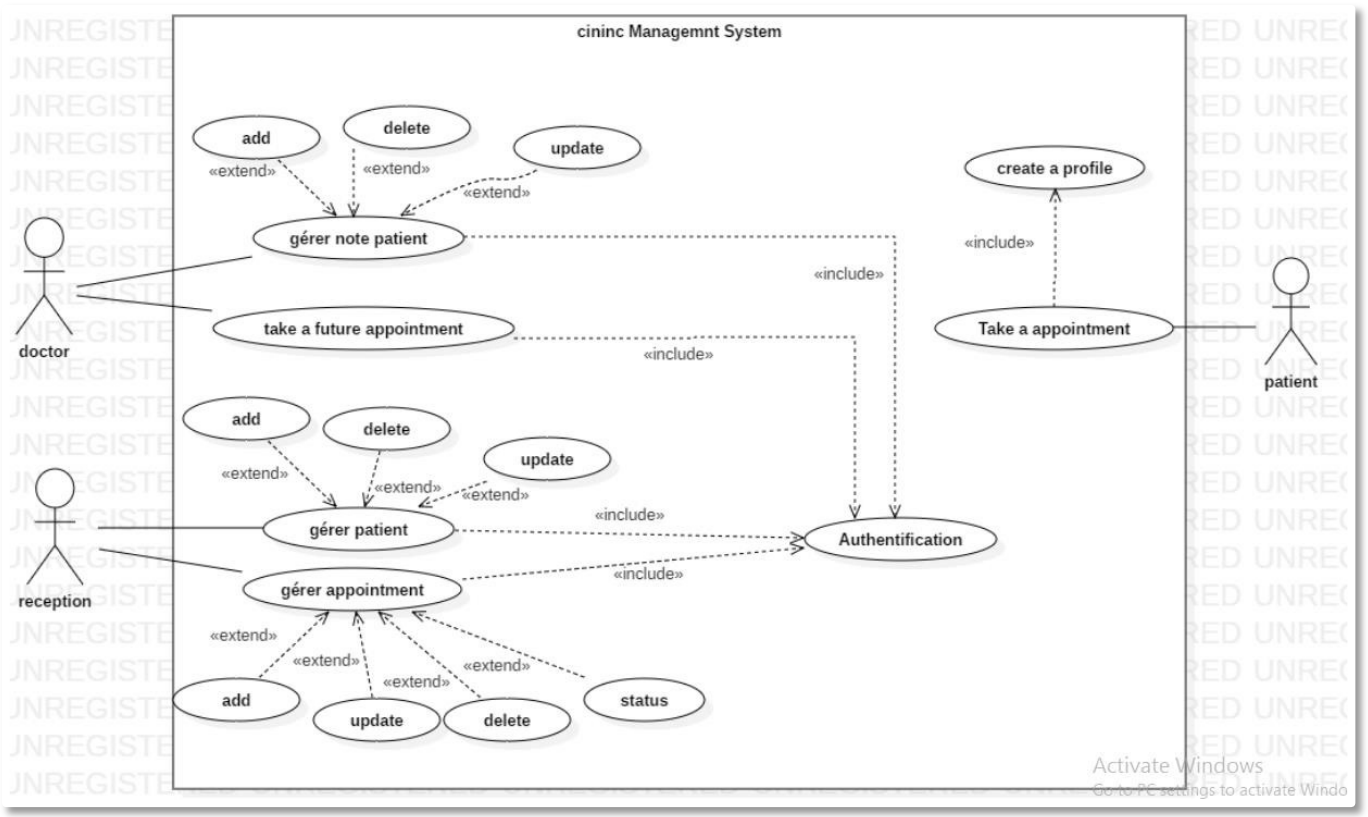


Figure 1: Use Case Diagram

I.4 Conclusion

In this chapter, we defined the requirements and presented the use case diagrams; in the next chapter, we will present the design of our application.

Chapter II: Design

II.1 Introduction

Design is a very important and essential step in the creation and development of software engineering applications; one of the most important design steps is modeling the various usage scenarios and the abstract structure of the application we will develop. To this end, we used UML (Unified Modeling Language).

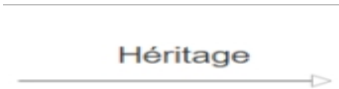
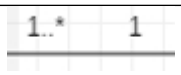
II.2 Class Diagrams

II.2.1 Definition

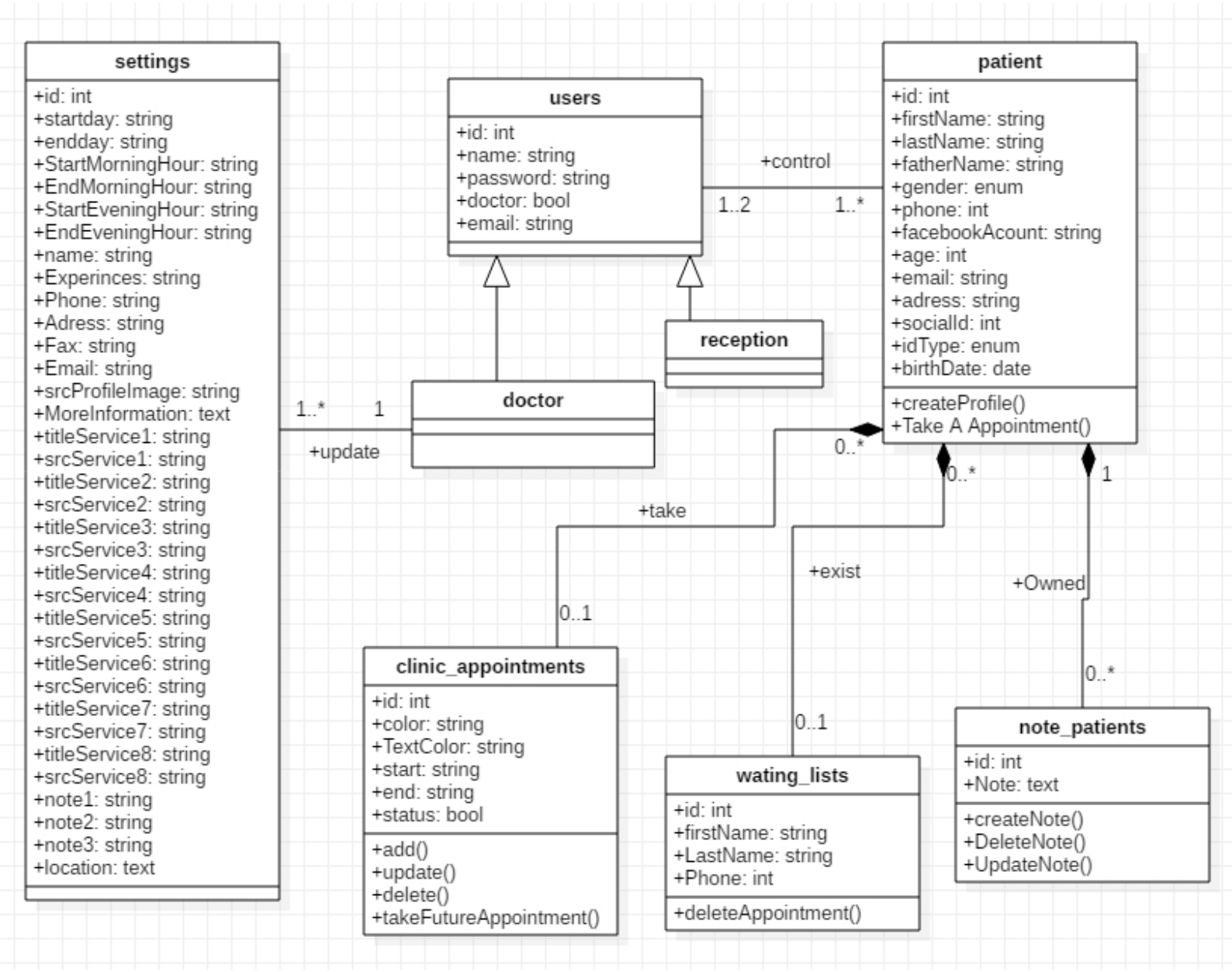
A class diagram is a diagram used in software engineering to represent the classes and interfaces of a system, as well as the various relationships between them.

This diagram is part of the static aspect of UML because it abstracts away temporal and dynamic aspects. [2]

II.2.2 Notations

Symbol	Meaning
+	**Public**
-	**Private**
"Type"	**Function return type**
Text	**Function or static variable**
	**Class**
	**Composition**
	**Inheritance**
	**Associations**

II.2.3 Class Diagram



II.3 Sequence Diagrams

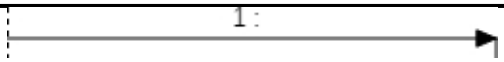
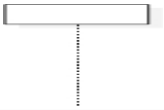
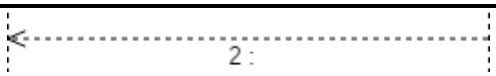
II.3.1 Definition

Sequence diagrams are graphical representations of interactions between actors and the system in chronological order, as defined by the Unified Modeling Language.

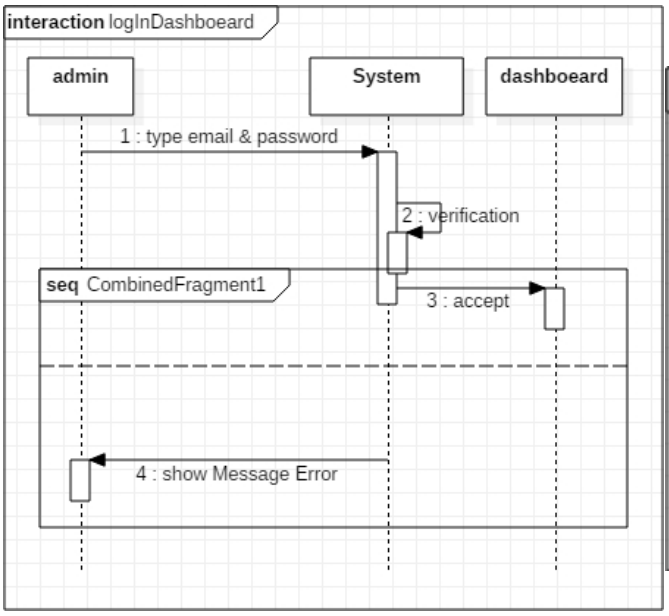
A sequence diagram is used to show the interactions between objects within the context of a use case scenario. For simplicity, the primary actor is depicted on the left side of the diagram, and any secondary actors are depicted on the right side of the system. The goal is to describe how actions unfold between actors or objects.

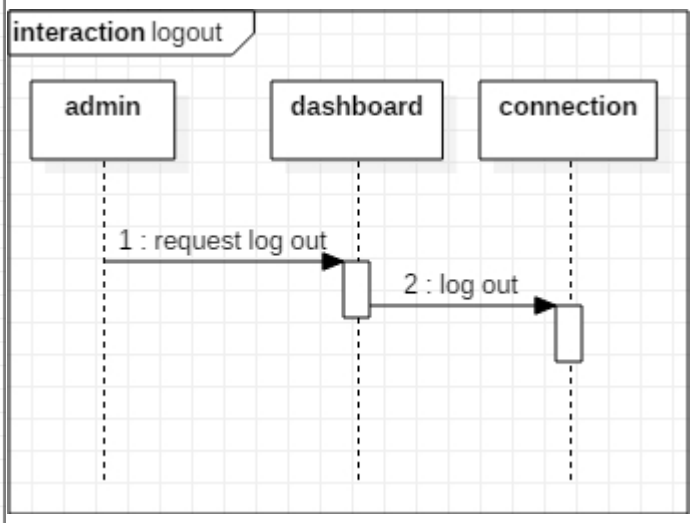
The vertical axis of the diagram represents time, allowing the sequence of actions over time to be visualized and the creation and destruction of objects to be specified. The periods of activity for objects are symbolized by rectangles, and these objects communicate via messages. [3]

II.3.2 Notations

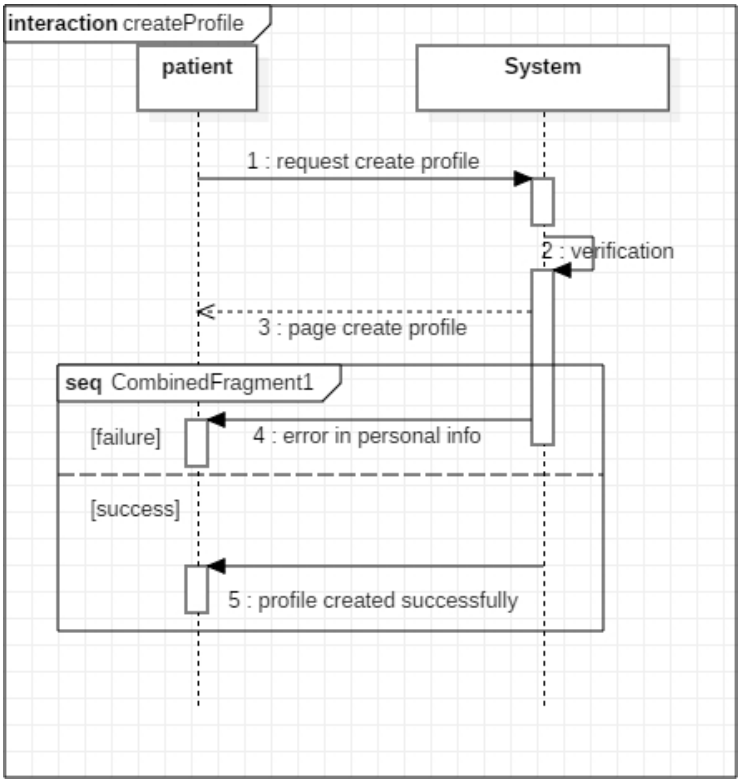
Symbol	**Meaning**
	Actor
	Message
	Timeline
	Feedback Message

II.3.3 Login/Logout Diagram

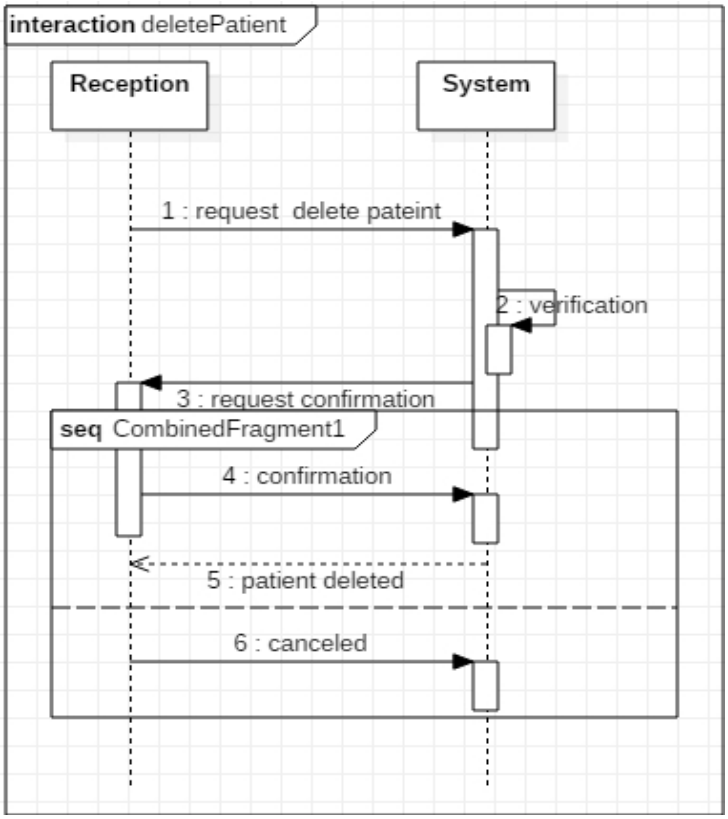




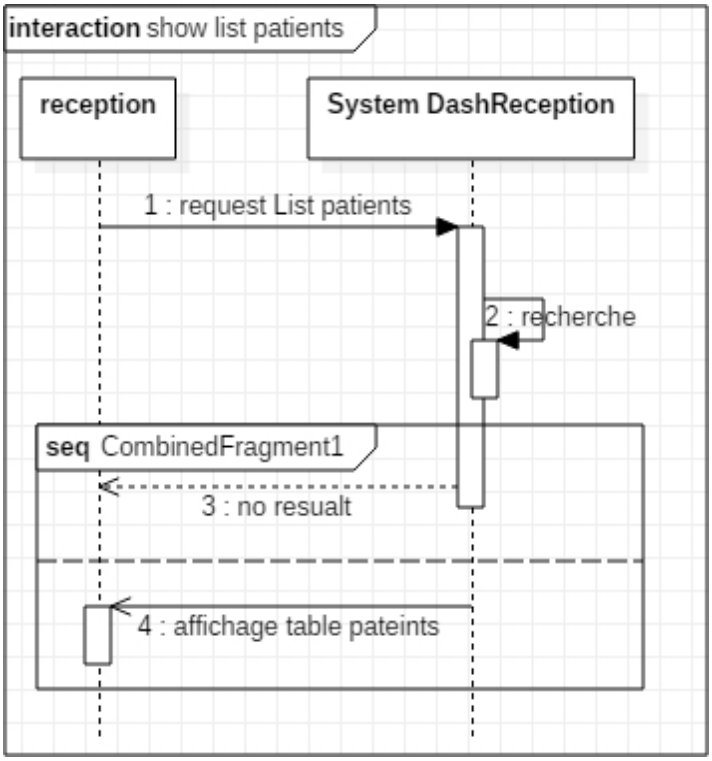
II.3.4 Create Profile Diagram



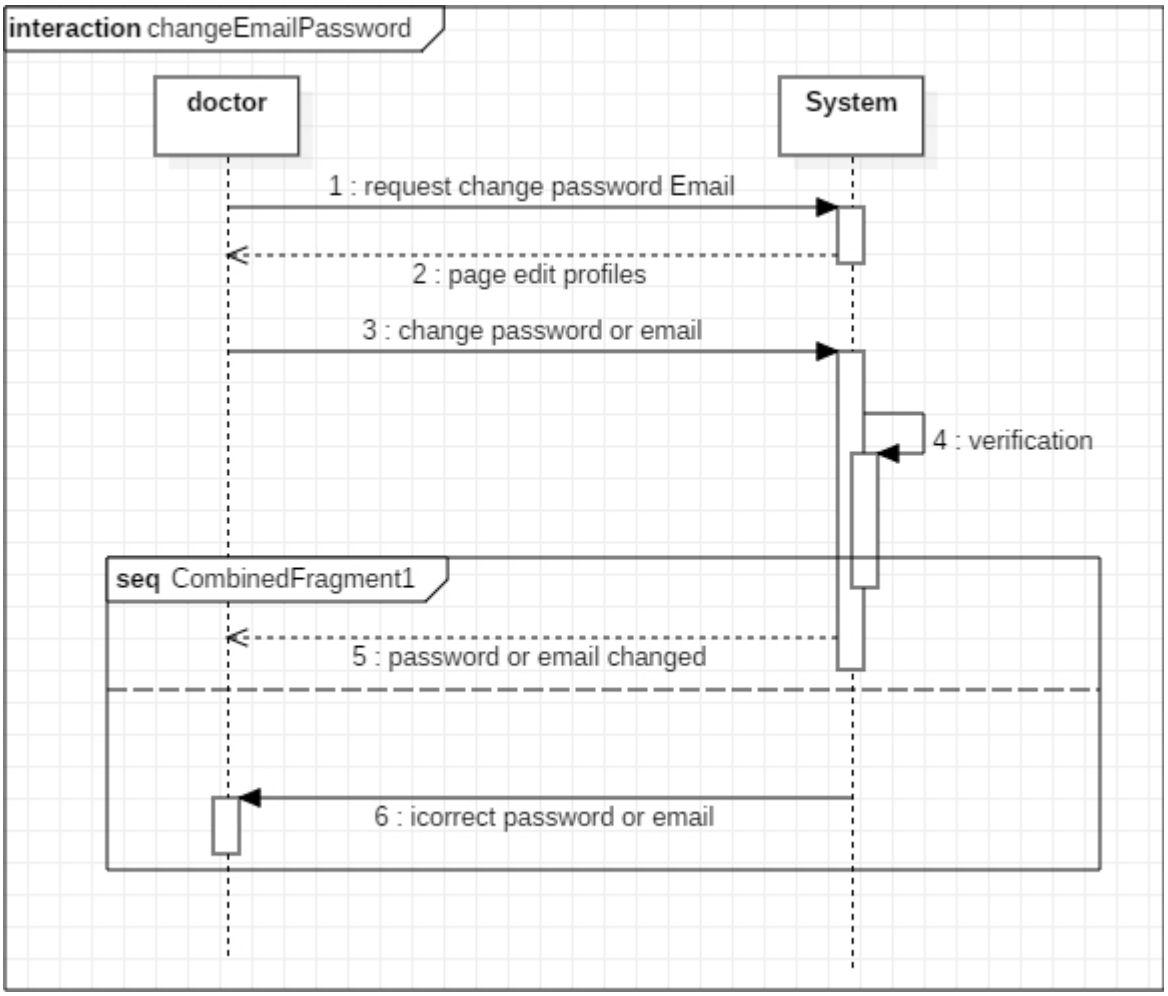
II.3.5 Delete Patient Diagram



II.3.6 Show Patients List Diagram



II.3.7 Password/Email Change Diagram



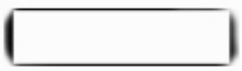
II.4 Activity Diagrams

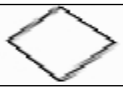
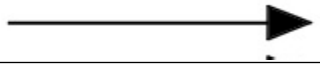
II.4.1 Definition

An activity diagram is a UML behavioral diagram used to represent the triggering of events based on the system's states and to model parallelizable behaviors (multithreaded or multiprocess).

The activity diagram is also used to describe a workflow. [4]

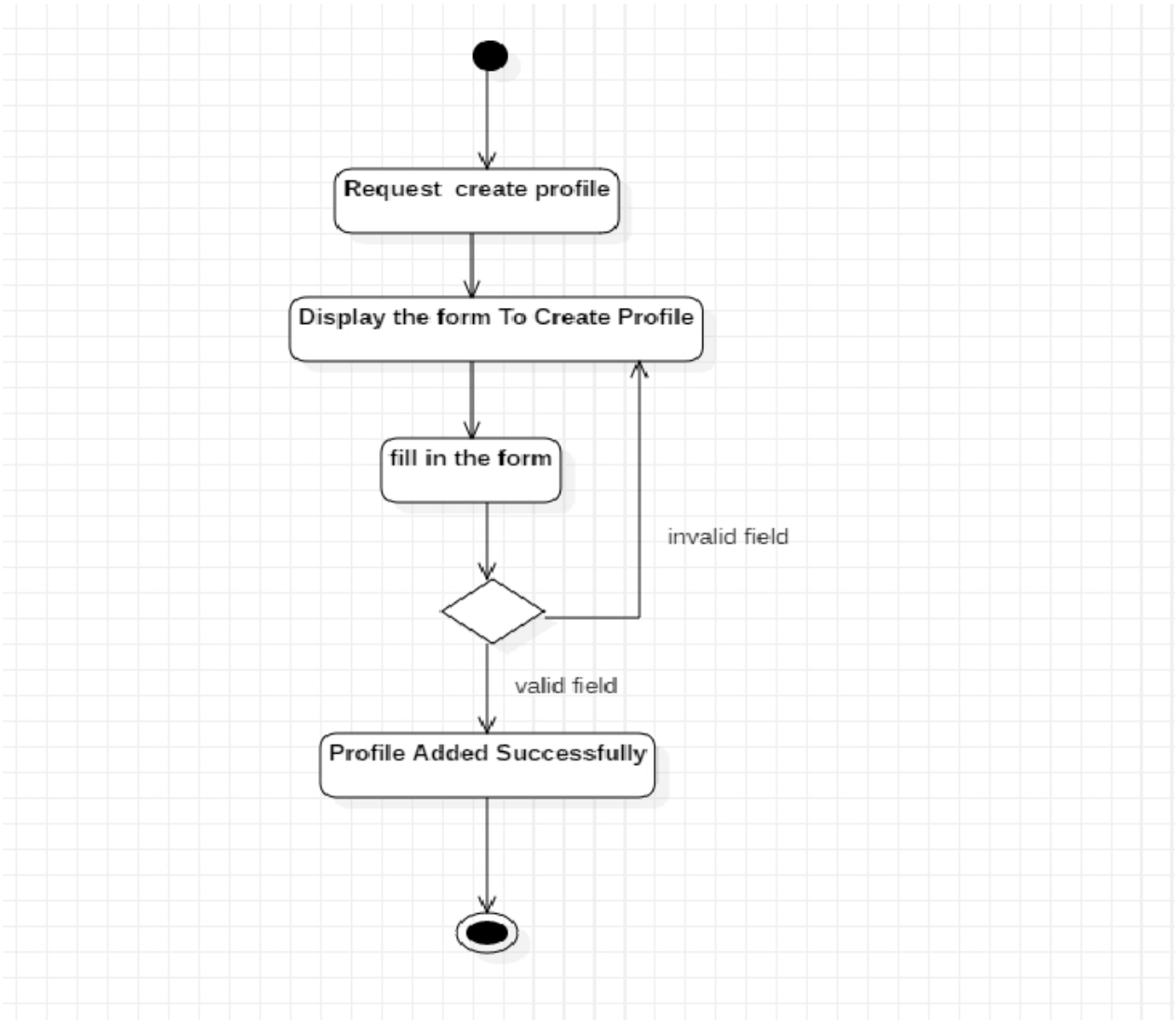
II.4.2 Notations

Symbols	Meanings
	Start state
	End state
	Action

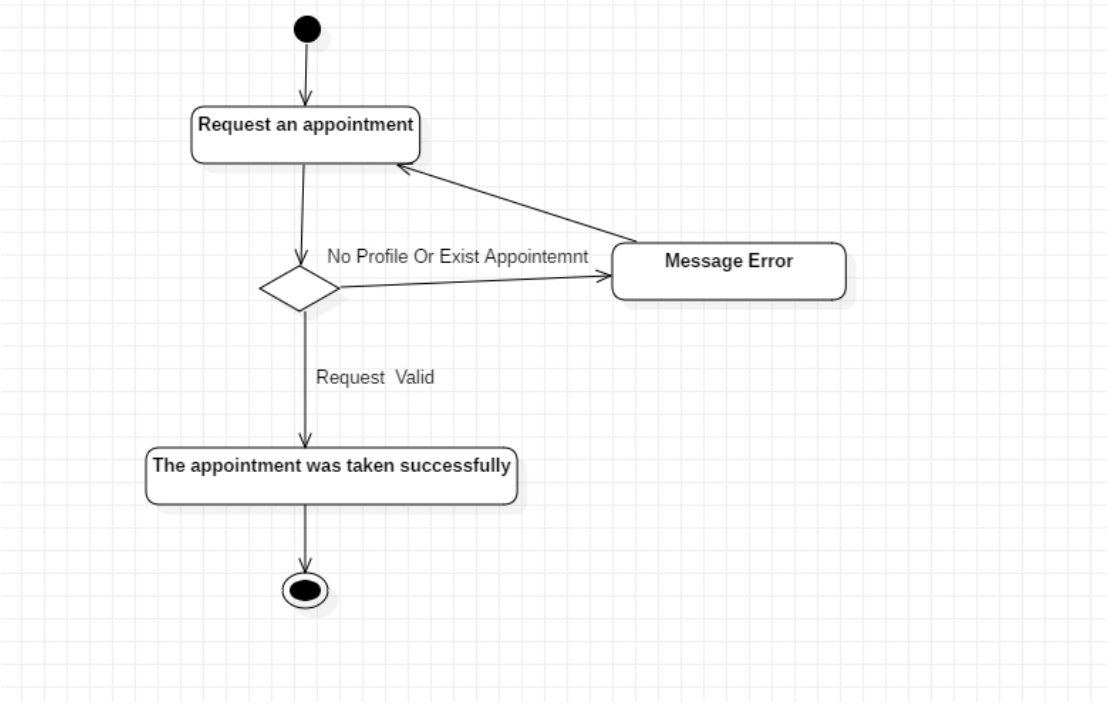
	Condition
	Control Arrow

II.4.3 Activity Diagram

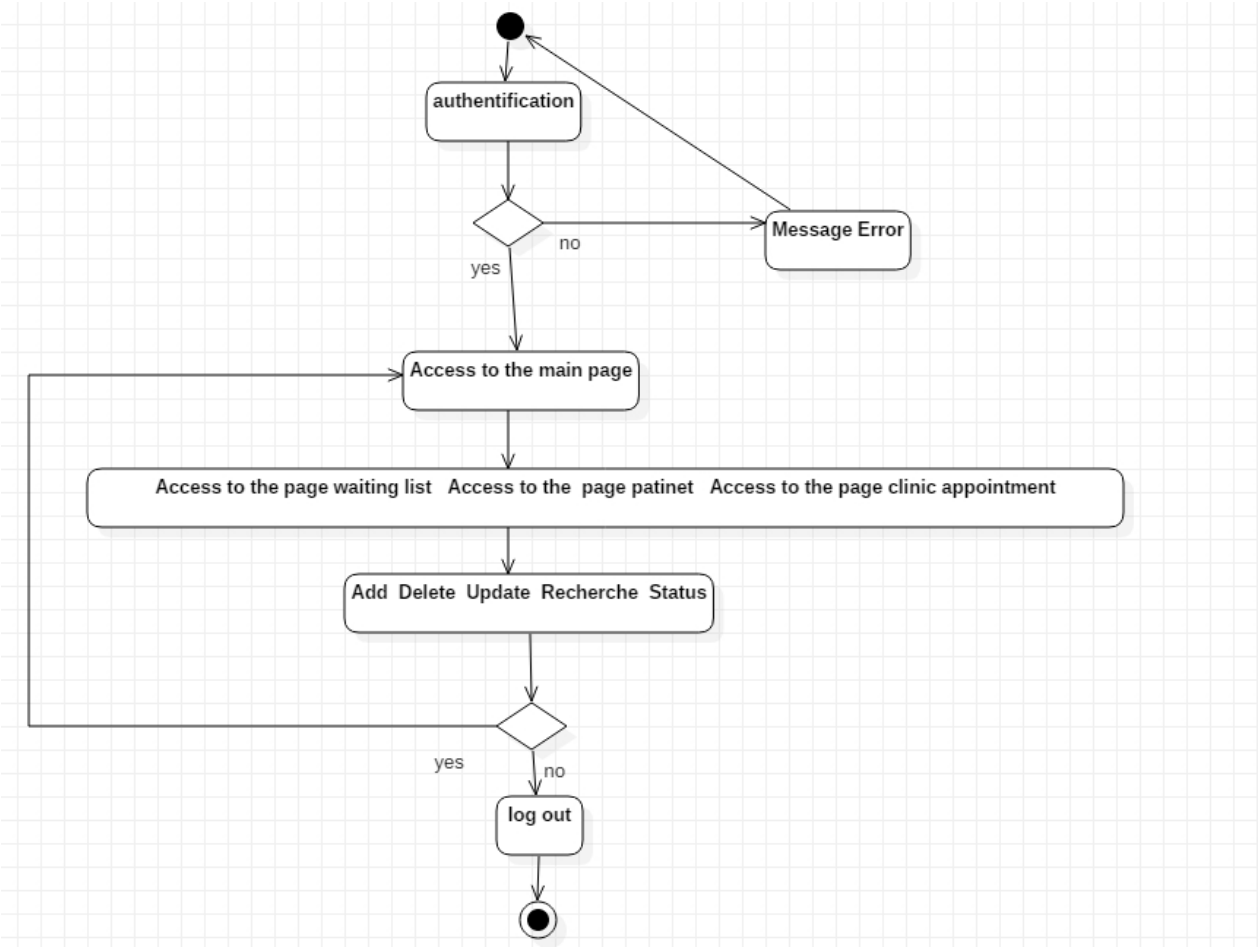
II.4.4 Profile Creation Diagram



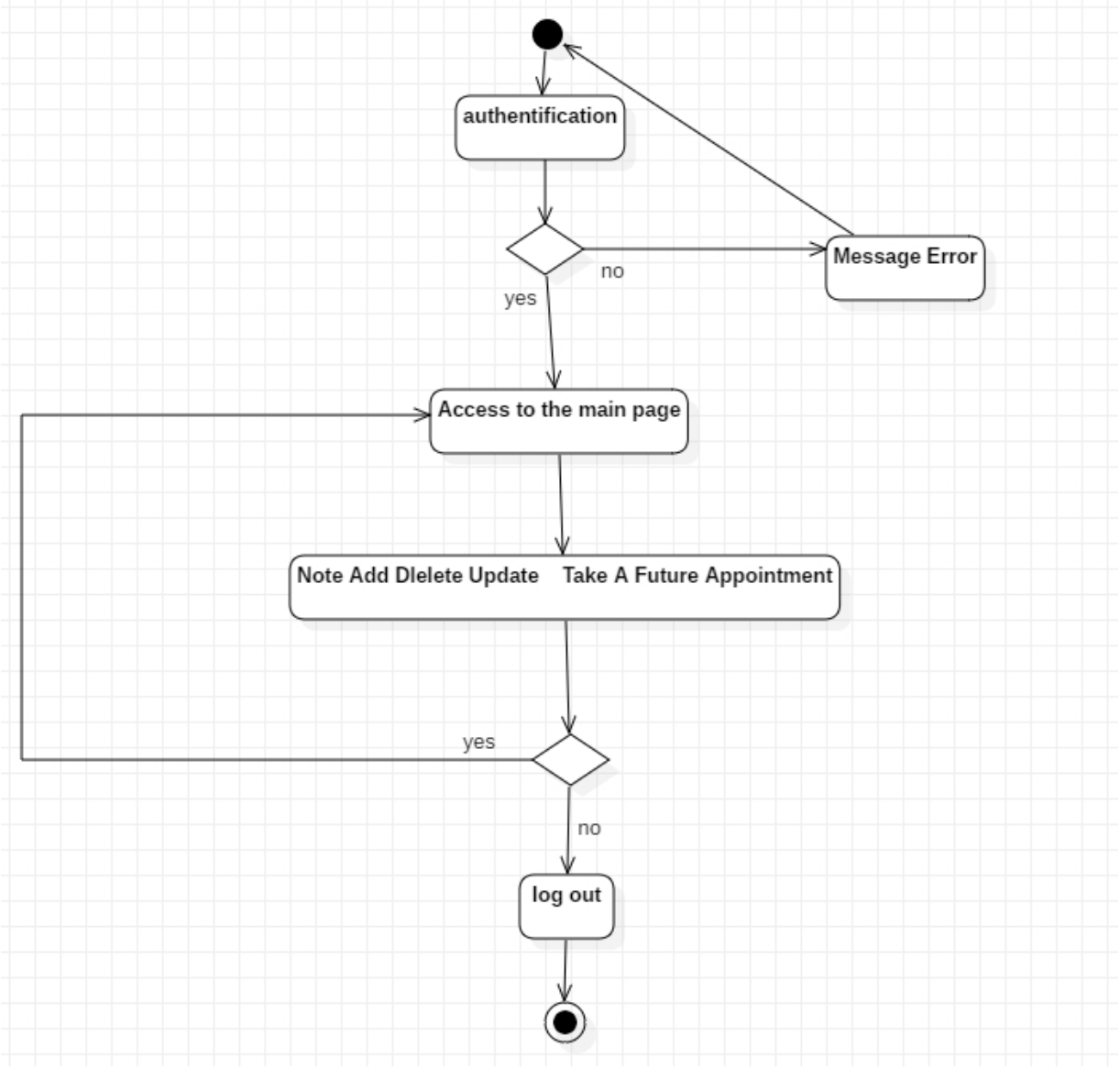
II.4.5 Schedule an Appointment Diagram



II.4.6 Reception Flowchart



II.4.7 Medical Flowchart



II.5 Conclusion

In the second chapter, we provided a conceptual explanation using UML, class diagrams, sequence diagrams, and activity diagrams.

We used these various diagrams to represent the different features of our application.

Chapter III : Implementation

III.1 Introduction

After analyzing and designing our application, we are beginning the development phase of this application, which uses technologies specific to these requirements.

In this chapter, we will first present the development environment and the languages used, and then move on to the various interfaces of our application.

III.2 Development Environment

III.2.1 Development Tools

III.2.1.1 XAMPP Server:

XAMPP is a software suite that makes it easy to set up a local web server, an FTP server, and an email server. It is an open-source software distribution (**X** (cross) Apache MariaDB Perl PHP) that offers great flexibility and is known for its quick and easy installation. As a result, it is accessible to a wide range of users since it does not require specialized knowledge and, moreover, runs on the most widely used operating systems.



It comes bundled with various software libraries that expand the range of services in a

Officially, XAMPP is intended for setting up a local test server before deploying a website, and its use is not recommended for a so-called *production* server.

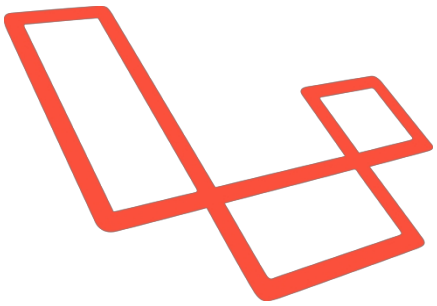
[5]

III.2.1.2 MySQL:

- **Overview of the MySQL Server**: MySQL is a robust and fast relational database management system (RDBMS). A database allows you to efficiently manipulate, sort, read, and search information. The MySQL server controls access to data to ensure that multiple users can simultaneously use the same database and access it quickly. MySQL is therefore a multi-user, multi-threaded server.
- **Characteristics of MySQL**: MySQL is a relational database server; it has the advantage of being geared toward serving existing data rather than toward frequent, secure updates. It is multithreaded, which it can handle multiple processes at the same time— and is also multi-user. [6]



III.2.1.3 Laravel:



Laravel is an open-source web framework written in PHP that follows the model-view-controller (MVC) pattern and is entirely developed using object-oriented programming. Laravel is distributed under the MIT license, with its source code hosted on GitHub. [7]

III.2.1.4 JavaScript:

JavaScript is a scripting language primarily used in interactive web pages but also for servers[2] through the use of (for example) Node.js[3]. It is a prototype-based, object-oriented language, meaning that the language's foundations and its main interfaces are provided by objects that are not



instances of classes, but each has constructors that allow their properties to be created, including a prototyping property that enables the creation of custom derived objects. Furthermore, functions are first-class objects. The language supports object-oriented, imperative, and functional programming paradigms. JavaScript has the largest ecosystem thanks to its dependency manager, npm, with approximately 500,000 packages as of August 2017.[4] [8]

III.2.1.5 Ajax:

The Ajax computing architecture (an acronym for Asynchronous JavaScript and XML) enables the creation of dynamic, interactive web applications and websites on the client side using various technologies added to web browsers between 1995 and 2005.

Ajax combines JavaScript, XMLHttpRequest-style requests, DOM manipulation, and a data format (XML or JSON) to improve the usability and user experience of rich web applications:

DOM and JavaScript allow you to modify the information displayed in the browser while preserving its structure;

The XMLHttpRequest object is used for asynchronous communication with the web server;

XML, as indicated by the acronym, was historically the preferred method for structuring information transmitted between the web server and the browser; today, JSON is increasingly replacing it for this purpose.

Ajax works on all common web browsers: Google Chrome, Safari, Mozilla Firefox, Internet Explorer, Microsoft Edge, Opera, etc. [9]

III.2.1.6 Bootstrap:

Bootstrap is a collection of tools useful for creating the design (graphics, animations, and interactions with the page in the browser, etc.) of websites and web applications. It is a framework that contains HTML and CSS code, forms, buttons, navigation tools, and other interactive elements, as well as optional JavaScript extensions. It is one of the most popular projects on the GitHub development management platform. [10]

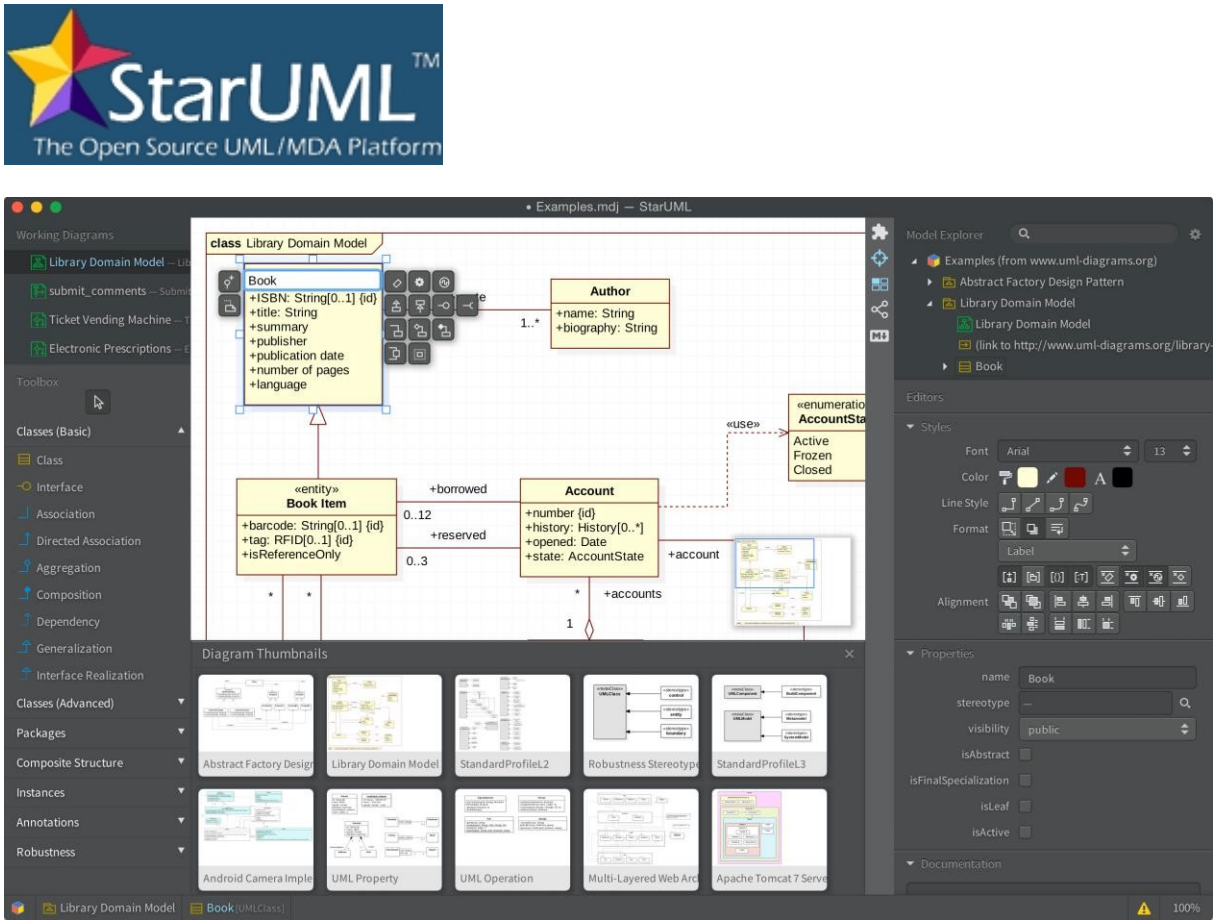
III.2.1.7 StarUML:

StarUML is UML modeling software that was released as open source by its publisher—after the end of its commercial operation—under a modified GNU GPL license.

The goal of taking over this project was to provide an alternative to commercial solutions such as IBM Rational Rose or Borland Together.

StarUML supports most of the diagrams specified in the UML 2.0 standard.

StarUML is written in Delphi 1 and relies on proprietary (non-open-source) Delphi components, which may explain why it is no longer being updated. [11]



III.2.1.8 SQL Query Language

SQL (Structured Query Language) can be considered the standardized language for accessing databases. It is now supported by most commercial products, whether by micro-database management systems such as Access or by more professional products such as Oracle. It has been the subject of several ANSI/ISO standards, the most widely used of which today is the SQL2 standard, which was defined in 1992.

The success of SQL is due primarily to its simplicity and the fact that it relies on the conceptual schema to formulate queries, leaving the DBMS responsible for the execution strategy. SQL provides a set-theoretic and declarative query language. However, SQL lacks the power of a full-fledged programming language—such as input/output operations, conditional statements, loops, and assignments. For certain tasks, it is therefore necessary to combine SQL with a more comprehensive programming language.

In summary, SQL is a relational language; it manipulates tables (i.e., relations, or sets) through queries that also produce tables. [12]

III.2.1.9 PHP:

- PHP is a scripting language most commonly used on the server side: in this architecture, the server interprets the PHP code in the requested web pages and generates code (such as HTML, XHTML, and CSS) and data (such as JPEG, GIF, and PNG) that can be interpreted and rendered by a web browser. PHP can also generate other formats such as WML, SVG, and PDF.
- It was designed to enable the creation of dynamic applications, most often developed for the Web. PHP is most commonly paired with an Apache server, although it can be installed on most HTTP servers such as IIS or nginx. This pairing allows for retrieving information from a database, a file system (file contents and directory structure), or simply data sent by the browser to be interpreted or stored for later use.
- It is a loosely typed and flexible language, making it easy for beginners to learn; however, this very flexibility can lead to security vulnerabilities in applications. Pragmatic by nature, PHP doesn't get bogged down in theory and tends to take the most direct approach. However, function names (as well as argument passing) do not always follow a consistent logic, which can make learning the language more challenging.
- Its use begins with form processing and then moves on to database access. Database access is straightforward once the corresponding modules have been installed on the server. The most obvious strength of this language is that, over time, it has made it easy to solve problems that were once complicated and has consequently become an essential component of web hosting packages.

- It is cross-platform: whether on Linux or Windows, it makes it easy to reuse the same code in a roughly similar environment (taking into account directory tree structures, which may vary).
- Open-source, free, easy to use and install, this language—like any programming language—requires a solid understanding of its core functions as well as a keen awareness of the security issues associated with it.
- Version 5.3 introduced many new features: namespaces—a fundamental element in the development of extensions, libraries, and
  structured frameworks, anonymous functions, closures, etc.
- In 2018, nearly 80% of websites used PHP in its various versions.[47]
- For several years now, the PHP language has been the focus of national gatherings organized by the AFUP[48] (the French Association of PHP Users), where programming experts and industry professionals come together to discuss PHP and its developers. The association organizes two major events[49]: the "PHP Forum," usually held at the end of the year, and "AFUP Day," which takes place during the first half of the year, simultaneously in several cities. [13]

III.3 Application Description

In this section, we will describe how our application works and the steps we followed during its development.

After launching the application, the following appears:

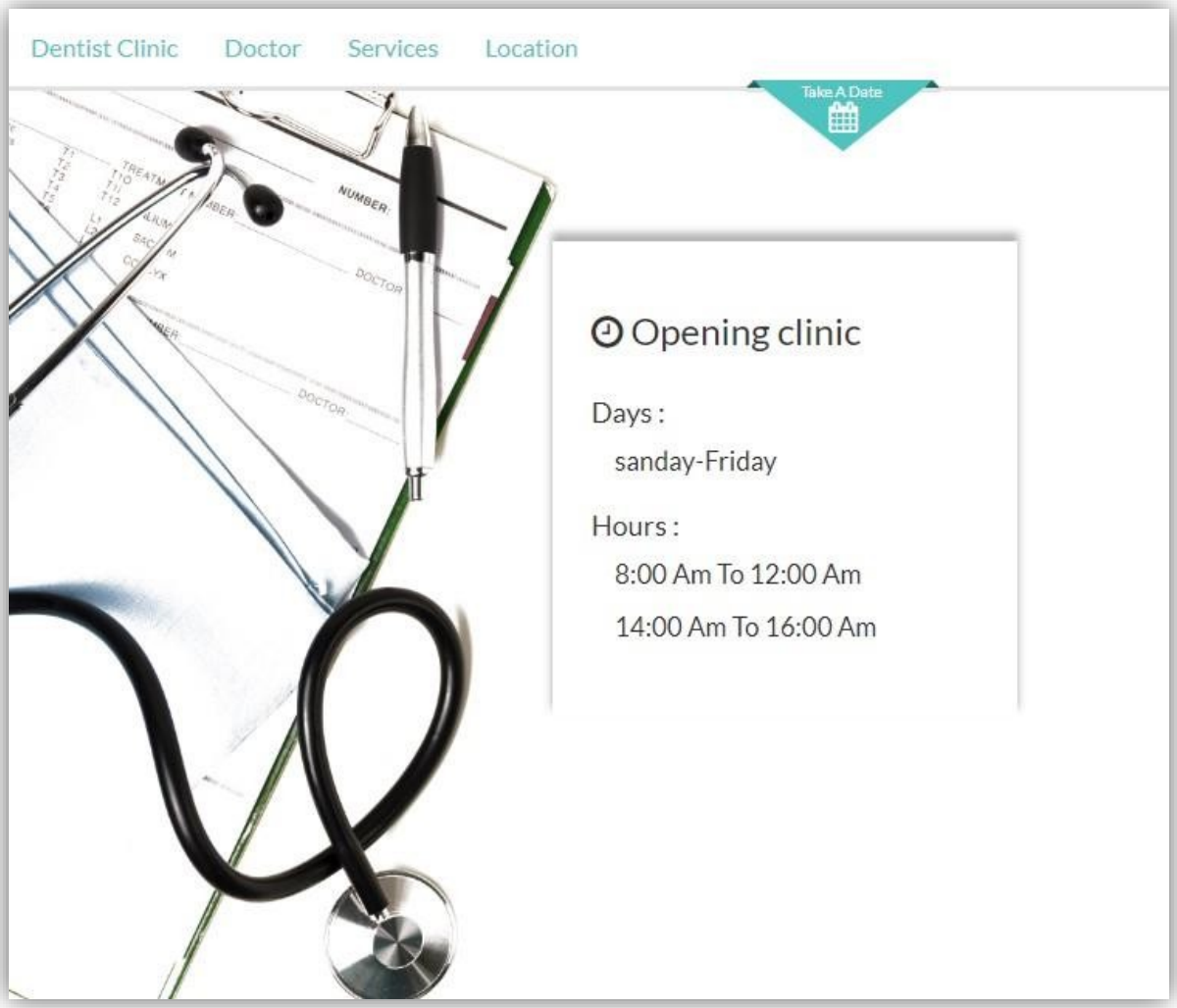


Figure 2: The application's home page

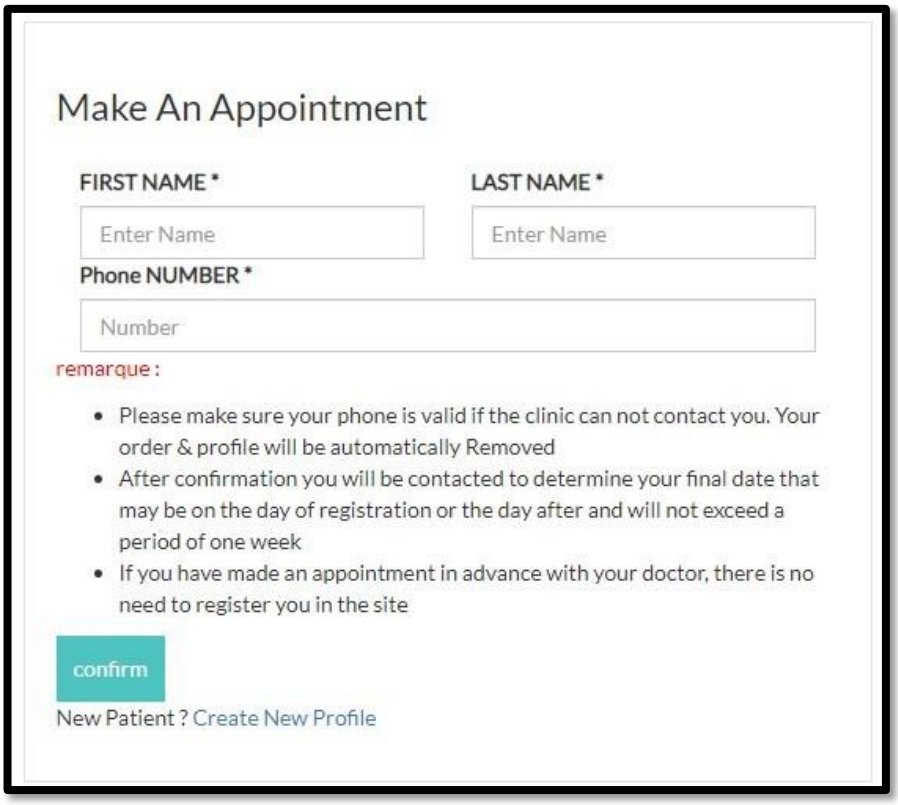


Figure 3: The patient schedules an appointment

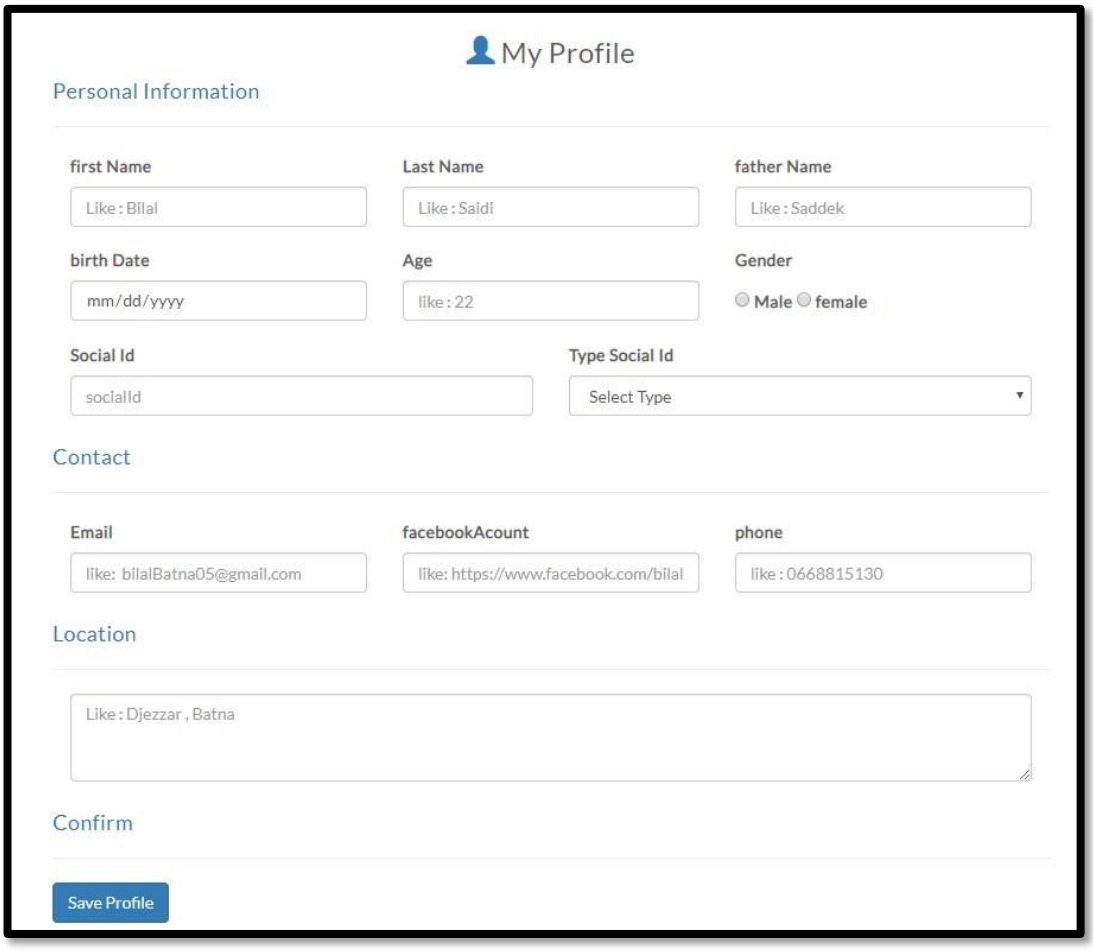


Figure 4: The patient creates a profile to confirm the appointment

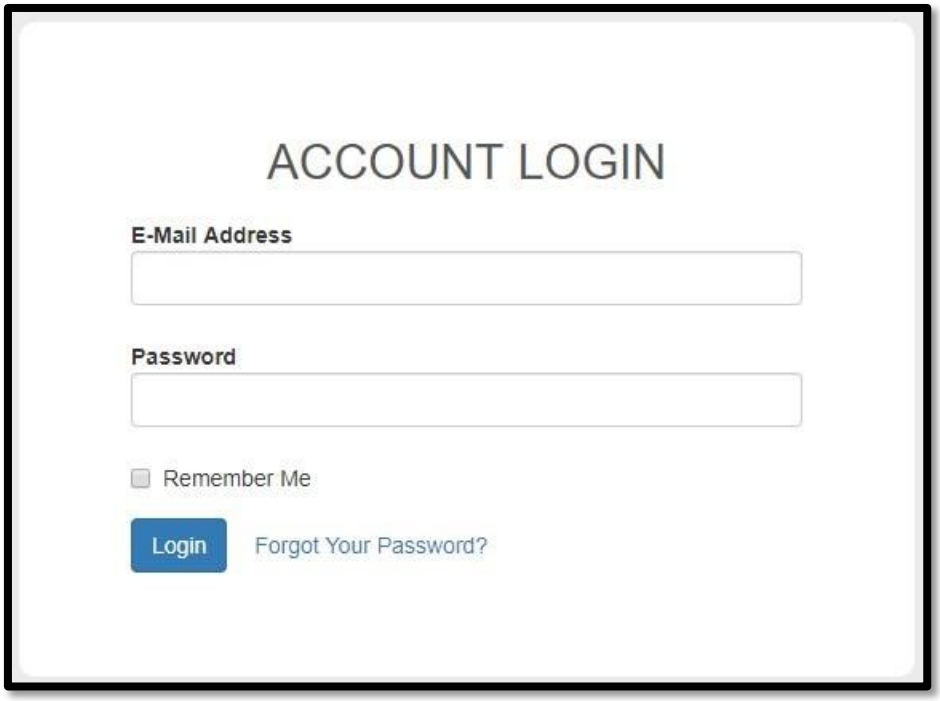


Figure 5: Admin authentication

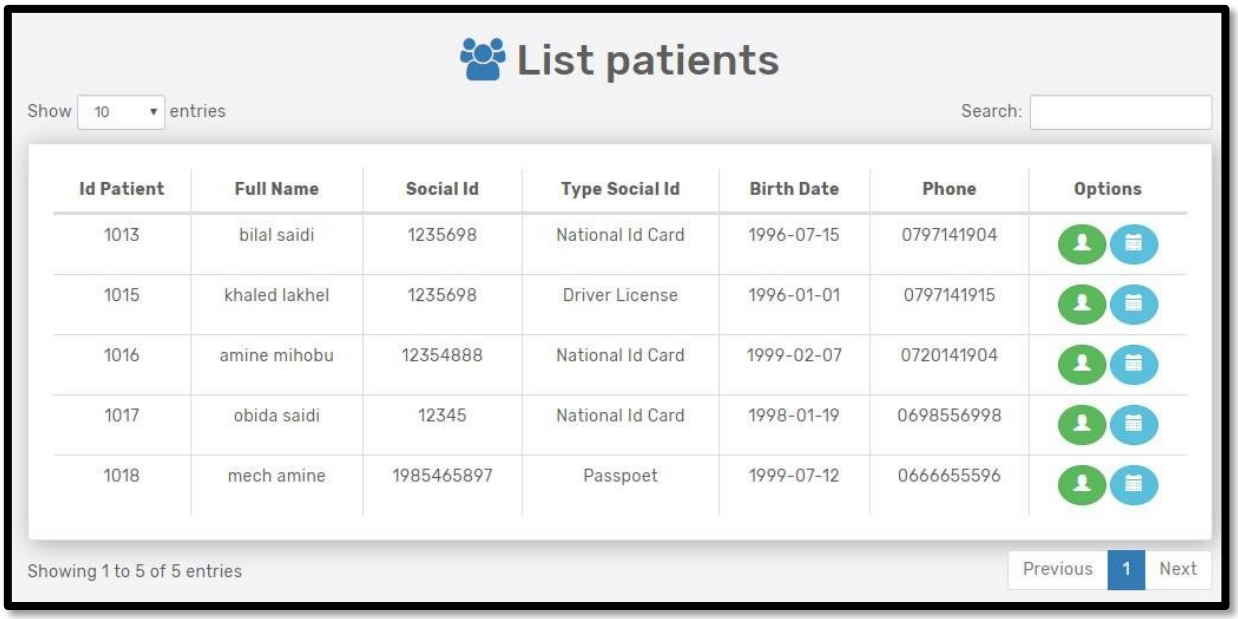


Figure 6: Reception Dashboard

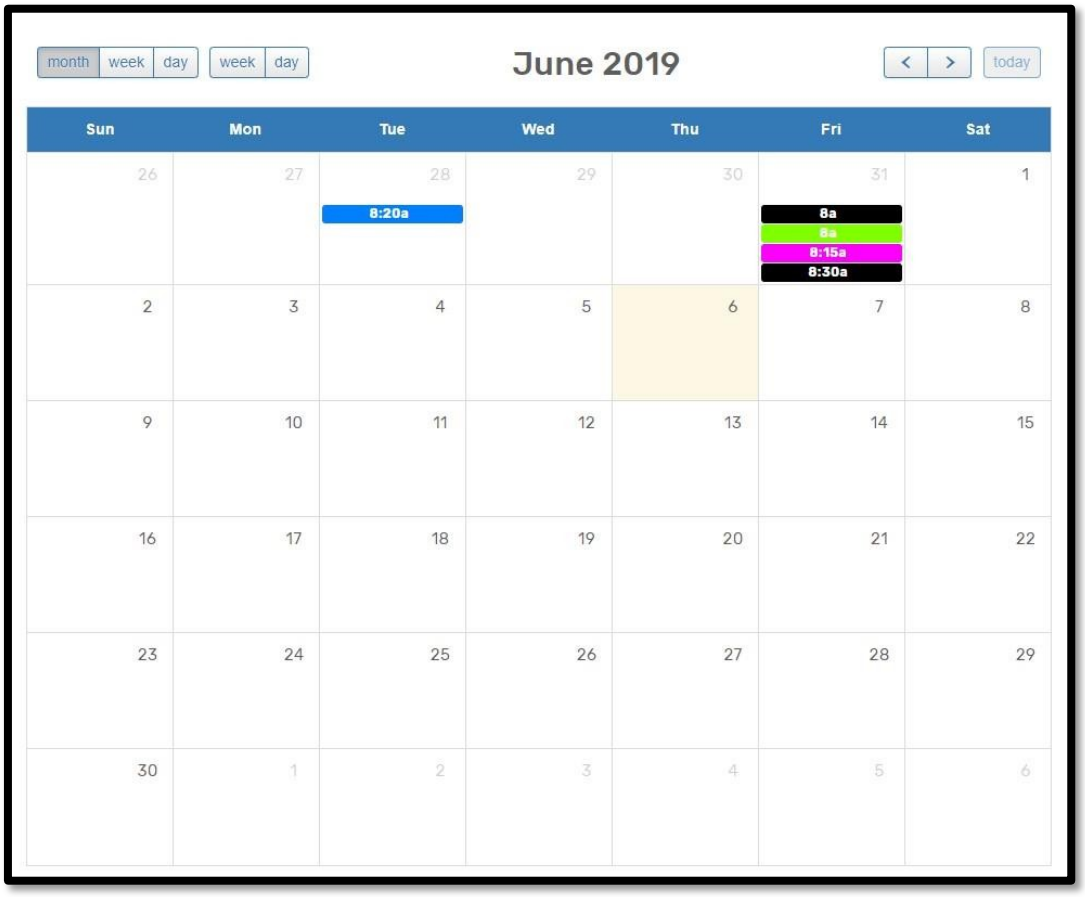


Figure 7: Calendar

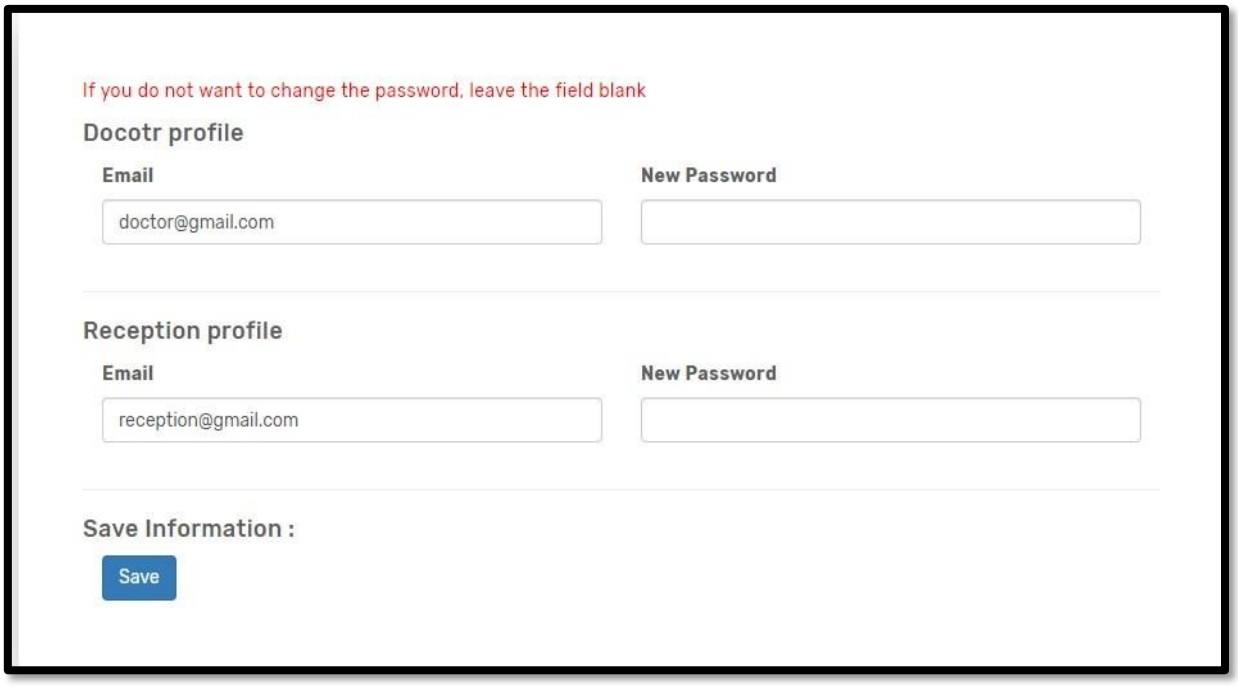


Figure 8: Administration Settings

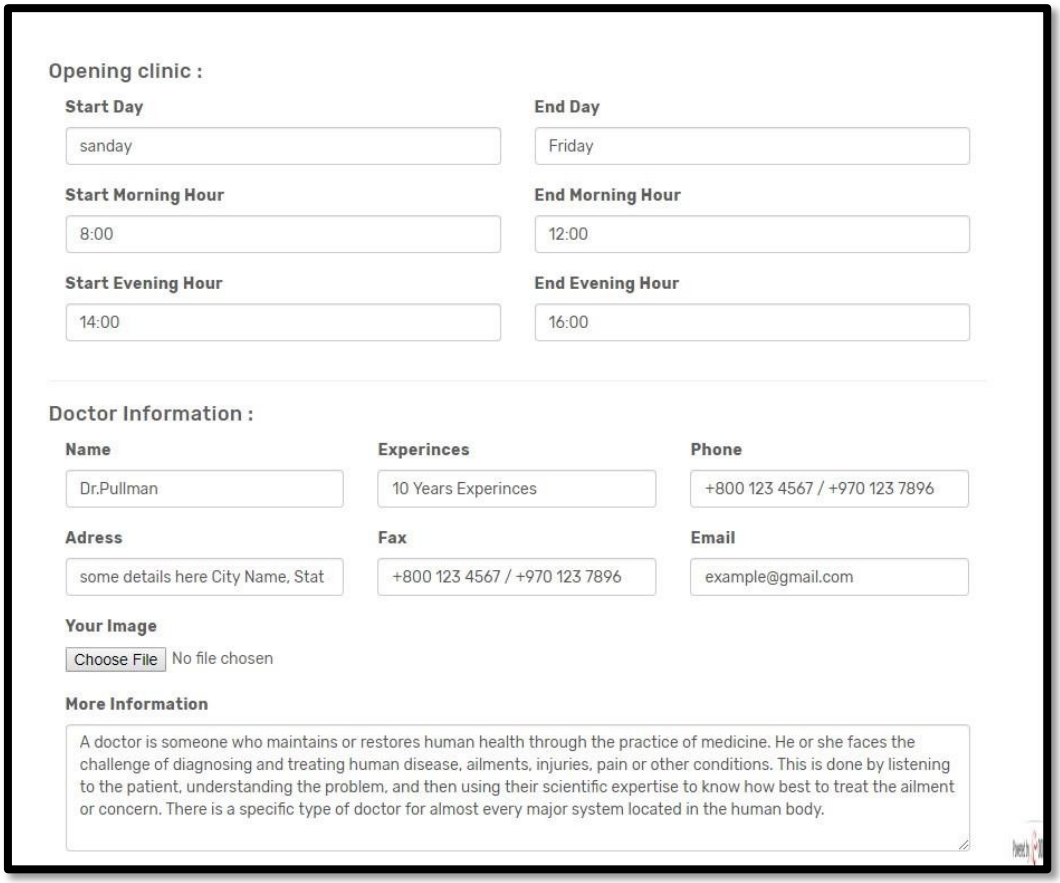


Figure 9: Clinic Settings



Figure 10: Patient waiting list



Figure 11: View Patient Profile

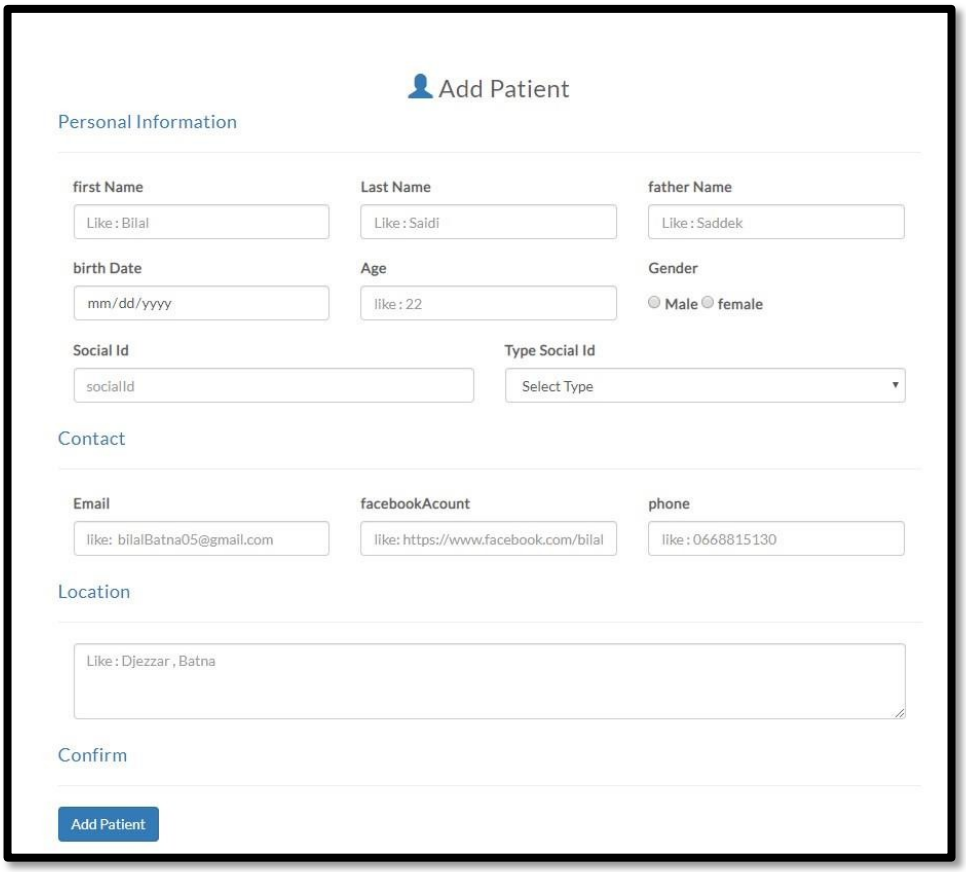


Figure 12: Add patient

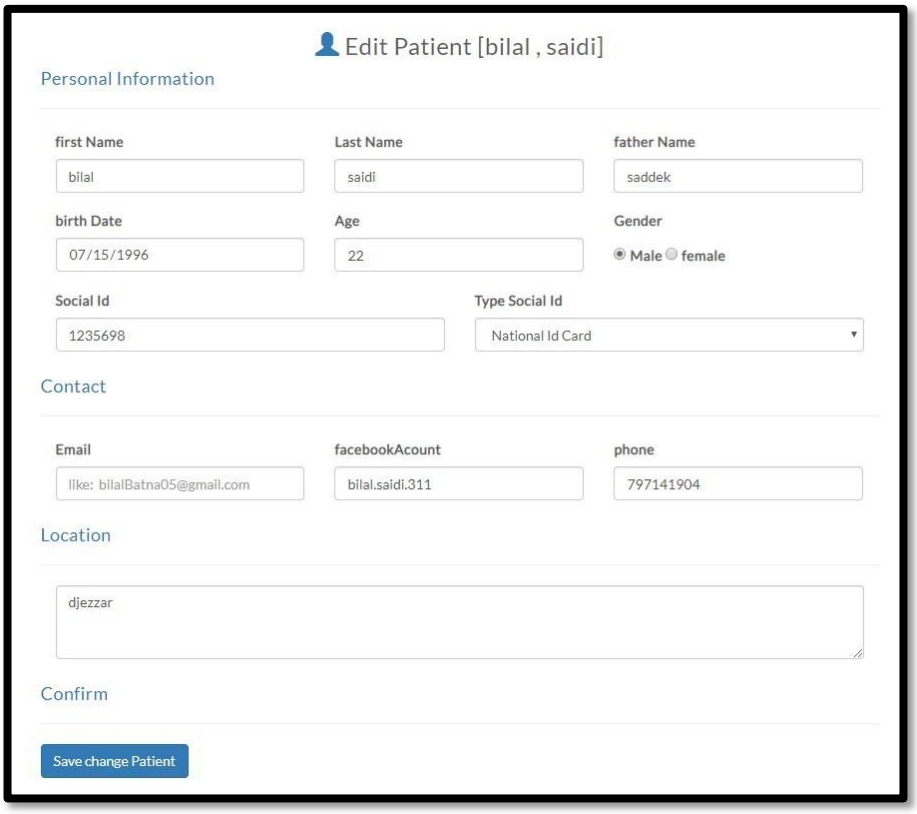


Figure 13: Edit patient

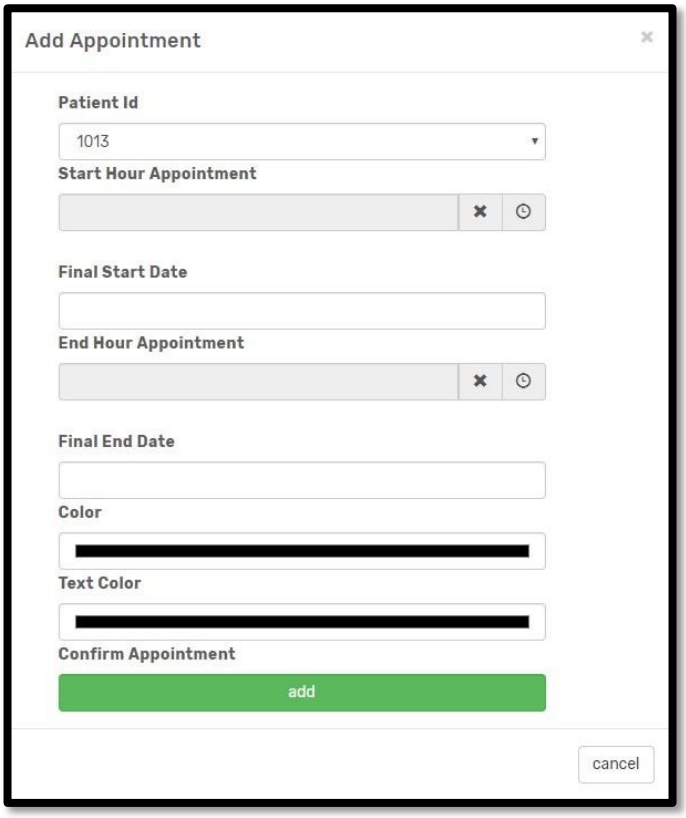


Figure 14: Add appointment

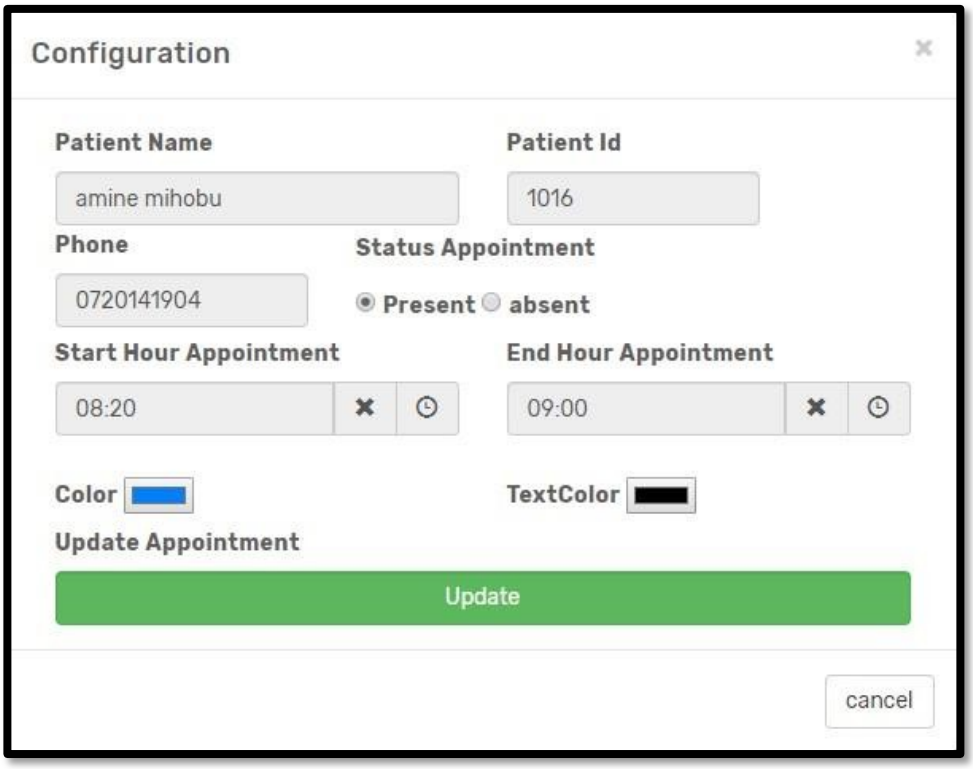


Figure 15: Edit appointment

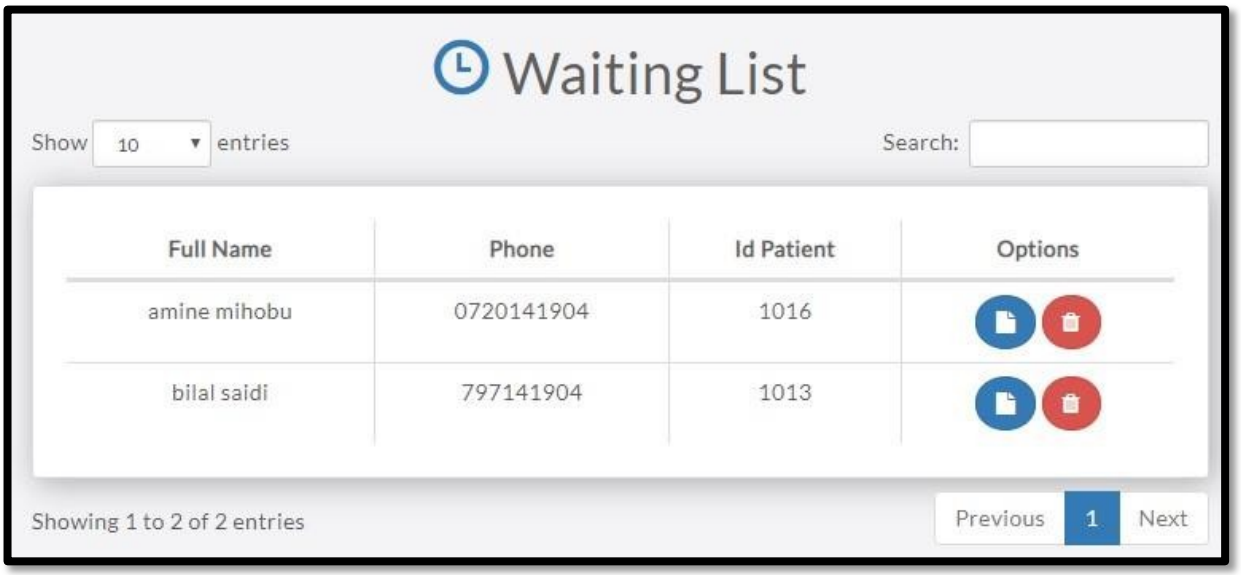


Figure 16: View or delete patient on the waiting list

Chapter IV General Conclusion

This final-year project involves developing a web-based medical management application.

In this thesis, we have presented the various stages of the design and implementation of our project.

Working on this project has taught us how to organize ourselves, how to collaborate, and has helped us improve our knowledge and skills in the field of programming. We have learned to better use the PHP language with the Laravel Framework and to effectively design websites using HTML/CSS/JS/Ajax.js.

In reality, since this modest project is a pilot effort, it is not a unique or perfect model; therefore, we are open to all criticism and ready to receive any suggestions and feedback aimed at further improving this initiative.

Note: The Editorial Team

- Keep it simple and clear
- Proofread your work (several times)
- Have someone else proofread your text (a third party—supervisor)
- Listen to others' feedback
- Don't hesitate to restructure

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Appendices